Stat 145 (Multivariate Statistics)

Lesson 3.4 Confirmatory Factor Analysis

Introduction

Recall that Exploratory Factor Analysis (EFA) is mainly used in a descriptive manner which sets it apart from inferential statistical methods. EFA is performed to identify latent constructs and generate hypotheses about structures between constructs, making EFA an exploratory, theory-generating method.

Meanwhile, the purpose of Confirmatory Factor Analysis is to evaluate structures of latent constructs that are hypothesized based on theories. CFA can be viewed as a form of structural equation modelling (SEM) as CFA directly corresponds to the measurement model in SEM. Similar to EFA, CFA is rooted in the common factor model and assumes that the relationships between indicators arise from effects of common latent constructs. In contrast to EFA, CFA requires researchers to specify the hypothesized characteristics of factor loadings and relationships between factors. This specification is typically linked to the tested theory and allows researchers to test hypotheses about the relationships between observed data and constructs, as well as the relationships between constructs. Besides assessing how well a specific model fits the data, CFA can also be employed to compare the fit of different models, or to determine which model provides the best fit to the observed data.

EFA and CFA can also be used together, but it is crucial to use different samples for the two procedures. It is not appropriate to conduct CFA in order to 'confirm' an EFA model using the same data. One way this can be circumvented is by considering disjoint subsamples for each procedure. Another use of EFA in conjunction with CFA is for exploring the underlying structure when the a priori specified model used in CFA does not fit the data.

Just like in EFA, we begin by testing the suitability of the CFA method. We do this again using two commonly used procedures: (1) Bartlett's test of sphericity, and (2) the Kaiser, Meyer, Olkin (KMO) test. Those tests are used to check whether a matrix is significantly different from an identity matrix. A significant Bartlett's test of sphericity states that FA may proceed if $p < 0.001$. In 1970, this method was modified by Kaiser as the Measure of Sampling Adequacy (MSA), and in 1974 by Kaiser and Rice. The KMO statistic can vary from 0 to 1 and indicates the degree to which each variable in a set is predicted without error by the other variables. The KMO statistic indicates how much variance in your variables might be caused by underlying factors. High values, close to 1.0, indicate that a factor analysis may be appropriate for your data.

Before running a model, the variables should be examined to check that there are no deviations from normality. The MVN package provides various approaches to check for univariate or multivariate deviation from a normal distribution. In practice, there are many cases where normality will not be met, especially in surveys using Likert scales. In such cases, we can use a robust estimator as presented in Finney and DiStefano (2013). In this lecture, we will use the maximum likelihood estimation with robust standard errors and a Satorra–Bentler scaled test statistic which can be used even in cases of deviations from the normality in most or all variables.

A simple exploratory visualization of the data can be performed to check for the correlations between the variables and to see the grouping of the variables in factors. It should be noted that correlation matrices produce different results from CFA when the relationships between the items are observed. Instead of using a correlation matrix, CFA uses a variance–covariance matrix on raw data to estimate input variance. Correlation graphs have been used to visualize relations and the grouping of variables at the same time. However, the most frequently used force-directed positioning of the network nodes representing variables is not easily interpretable and can be misleading when grouping of the variables is observed. Therefore, alternative options are available in the qgraph package.

First, the factor loadings of the indicators (observed variables) that make up the latent construct need to be calculated. The standardized factor loading squared is the estimate of the amount of the variance of the indicator that is accounted for by the latent construct. Factor loadings of 0.4 or higher are acceptable (Hair et al., 2010).

The unique variance is not explained by the latent construct. We also need to check the convergent validity of the construct, which is indicated by high indicator loadings, which shows the strength of how well the indicators are theoretically similar. Most models contain more than one factor. In that case, we need to run a CFA for all the model's latent constructs within one measurement model. Discriminant validity exists when no two constructs are highly correlated. If two constructs are highly correlated (> 0.85), we need to explore combining the constructs.

Many different approaches are proposed to assess the fit of the model to the data (Jackson et al., 2009). Some of the more popular fit statistics include the comparative fit index (CFI), the Tucker–Lewis index (TLI), and root mean square error of approximation (RMSEA). CFI is frequently used as it is known to perform well even when the sample size is small (Tabachnick & Fidell, 2007) and assumes that all latent variables are uncorrelated and compares the postulated model to the null model. The CFI values can range from 0.0 to 1.0, where values close to 1.0 represent a good fit. To avoid misspecified models it is generally advised for models to achieve CFI over 0.90 which represents a good fit with some authors arguing that the threshold should lie at 0.95 (Hu & Bentler, 1999). TLI measures a relative reduction in misfit per degree of freedom (Tucker & Lewis, 1973) with threshold values of 0.90 indicating adequate fit and 0.95 indicating good fit (Qian et al., 2023). On the other hand, the RMSEA represents a so-called badnessof-fit measure resulting in lower values for a better fit. It measures discrepancy due to the approximation with values below 0.06 representing an acceptable model (Shi et al., 2020).

Implementing CFA in R

```r
#Loading packages
library(psych)
library(lavaan)
library(semPlot)
library(smacof)
library(parameters)
library(performance)
library(MVN)
library(semTools)
```

The data

We used the dataset originally from the study by Snowden et al. (2015) to examine the construct validity of the Trait Emotional Intelligence Questionnaire Short form (TEIQue-SF). The TEIQue-SF is a short-form questionnaire which includes 30 items. It was developed by Petrides (2009) to measure global trait emotional intelligence (trait EI). Petrides derived it from the larger 130-item-based TEIQue questionnaire by Freudenthaler et al. (2008). Items were selected based on their correlations with the corresponding total facet scores (Freudenthaler et al., 2008). Responses to the TEIQue-SF items are made on a Likert-type scale (e.g., 1 meaning "strongly disagree" and 7 meaning "strongly agree"). The total scale score is derived by summing the score of each item (after reverse scoring of negative items) and is used to locate respondents on the latent trait continuum. A higher score represents a greater presence of the trait EI [20]. Four dimensions of trait EI are measured with the TEIQue-SF: well-being, sociability, self-control, and emotionality. Petrides (2009) defines well-being as related feelings of a time based around achievements, self-regard, and expectations; self-control as regulating and having control over emotions, impulses, and stress; emotionality as an ability to perceive, express, and connect with emotions in self and others, which can be used in creating successful interpersonal relationships; and sociability as being socially assertive and aware, managing others' emotions, and effectiveness in communication and participation in social situations.

```r
#Loading the data
TEIQ <- read.csv("db_TEIQ_CFA_SCO.csv", header = T)
#head(TEIQ)
```

Items 3, 18, 14 & 29 were omitted from the CFA because Petrides (2006) considered them a 'general' factor as they were not specifically associated with any particular factor. Additionally, item number 20 was omitted as suggested by Snowden et al. (2016). Finally, the columns were rearranged to match four subscales.

```r
#Data preparations
teiq <- TEIQ[,2:31]
colnames(teiq) <- paste0("V", 1:30)
teiq <- teiq[,-c(3, 14, 18, 20, 29)]
items <- c("V30","V15","V19","V24","V27","V21","V9","V6", #Confidence
           "V12","V5","V28","V13","V16", #Connection
           "V7","V10","V22","V25","V8","V4","V2", #Uncertainty
           "V11","V26","V17","V1","V23") #Empathy
pos <- match(items, colnames(teiq))
teiq <- teiq[,items]
#head(teiq)
```

Testing the MVN assumption

```r
mvn.out <- mvn(data = teiq,
               mvn_test = "hz",
               univariate_test = "SW",
               descriptives = TRUE)
mvn.out$univariate_normality
```

```
##             Test Variable Statistic p.value    Normality
## 1  Shapiro-Wilk      V30     0.939  <0.001   Not normal
## 2  Shapiro-Wilk      V15     0.920  <0.001   Not normal
## 3  Shapiro-Wilk      V19     0.910  <0.001   Not normal
## 4  Shapiro-Wilk      V24     0.925  <0.001   Not normal
## 5  Shapiro-Wilk      V27     0.890  <0.001   Not normal
## 6  Shapiro-Wilk      V21     0.918  <0.001   Not normal
## 7  Shapiro-Wilk       V9     0.849  <0.001   Not normal
## 8  Shapiro-Wilk       V6     0.837  <0.001   Not normal
## 9  Shapiro-Wilk      V12     0.683  <0.001   Not normal
## 10 Shapiro-Wilk       V5     0.613  <0.001   Not normal
## 11 Shapiro-Wilk      V28     0.718  <0.001   Not normal
## 12 Shapiro-Wilk      V13     0.528  <0.001   Not normal
## 13 Shapiro-Wilk      V16     0.810  <0.001   Not normal
## 14 Shapiro-Wilk       V7     0.942  <0.001   Not normal
## 15 Shapiro-Wilk      V10     0.911  <0.001   Not normal
## 16 Shapiro-Wilk      V22     0.925  <0.001   Not normal
## 17 Shapiro-Wilk      V25     0.933  <0.001   Not normal
## 18 Shapiro-Wilk       V8     0.847  <0.001   Not normal
## 19 Shapiro-Wilk       V4     0.917  <0.001   Not normal
## 20 Shapiro-Wilk       V2     0.826  <0.001   Not normal
## 21 Shapiro-Wilk      V11     0.934  <0.001   Not normal
```

```
## 22 Shapiro-Wilk      V26     0.925  <0.001   Not normal
## 23 Shapiro-Wilk      V17     0.917  <0.001   Not normal
## 24 Shapiro-Wilk       V1     0.922  <0.001   Not normal
## 25 Shapiro-Wilk      V23     0.944  <0.001   Not normal
```

```
mvn.out$multivariate_normality
```

```
##            Test Statistic p.value     Method          MVN
## 1 Henze-Zirkler     1.048  <0.001 asymptotic   Not normal
```

None of the items is normally distributed and consequently, the multivariate normality test shows the same. Since we are using Likert-scale items, this is not unusual.

Testing suitability of data for CFA

```
# KMO test
KMO(teiq)
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: KMO(r = teiq)
## Overall MSA =  0.88
## MSA for each item =
##  V30  V15  V19  V24  V27  V21   V9   V6  V12   V5  V28  V13  V16   V7  V10  V22
## 0.85 0.91 0.89 0.87 0.91 0.92 0.90 0.91 0.86 0.86 0.91 0.88 0.85 0.89 0.82 0.89
##  V25   V8   V4   V2  V11  V26  V17   V1  V23
## 0.76 0.91 0.91 0.85 0.69 0.83 0.83 0.90 0.78
```

```
#Bartlett's test
cortest.bartlett(teiq)
```

```
## $chisq
## [1] 4841.025
##
## $p.value
## [1] 0
##
## $df
## [1] 300
```

The KMO MSA suggests that the data are appropriate for FA ($KMO = 0.88$). Bartlett's test of sphericity suggests that there is sufficient significant correlation in the data for FA ($\chi^2_{(300)} = 4841.03, p < 0.001$).

Since the data are not normally distributed, a robust estimator following Finney and DiStefano (2013) will be used to perform CFA. More specifically, we will use maximum likelihood estimation with robust standard errors and a Satorra-Bentler scaled test statistic.

Building the CFA model

```
cfa.model <- 'Confidence =~ V30 + V15 + V19 + V24 + V27 + V21 + V9 + V6
          Connection =~ V12 + V5 + V28 + V13 + V16
          Uncertainty =~ V7 + V10 + V22 + V25 + V8 + V4 + V2
          Empathy =~ V11 + V26 + V17 + V1 + V23'
```

Examining the model fit statistics

```
model.fit <- cfa(model = cfa.model,
                 data = teiq,
                 estimator = "MLM",
                 std.lv=TRUE)

# Print model summary
summary(model.fit, fit.measures=TRUE, standardized=TRUE)
```

```
## lavaan 0.6-20 ended normally after 25 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        56
##
##   Number of observations                           938
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              1105.117     926.518
##   Degrees of freedom                               269         269
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.193
##     Satorra-Bentler correction
##
```

```
## Model Test Baseline Model:
##
##   Test statistic                               4894.071    3895.527
##   Degrees of freedom                                300         300
##   P-value                                         0.000       0.000
##   Scaling correction factor                                   1.256
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                     0.818       0.817
##   Tucker-Lewis Index (TLI)                        0.797       0.796
##
##   Robust Comparative Fit Index (CFI)                          0.826
##   Robust Tucker-Lewis Index (TLI)                             0.806
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -38833.831  -38833.831
##   Loglikelihood unrestricted model (H1)      -38281.272  -38281.272
##
##   Akaike (AIC)                                77779.661   77779.661
##   Bayesian (BIC)                              78050.911   78050.911
##   Sample-size adjusted Bayesian (SABIC)       77873.059   77873.059
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                           0.058       0.051
##   90 Percent confidence interval - lower          0.054       0.048
##   90 Percent confidence interval - upper          0.061       0.054
##   P-value H_0: RMSEA <= 0.050                     0.000       0.295
##   P-value H_0: RMSEA >= 0.080                     0.000       0.000
##
##   Robust RMSEA                                                0.056
##   90 Percent confidence interval - lower                      0.052
##   90 Percent confidence interval - upper                      0.060
##   P-value H_0: Robust RMSEA <= 0.050                          0.008
##   P-value H_0: Robust RMSEA >= 0.080                          0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                            0.056       0.056
##
## Parameter Estimates:
##
##   Standard errors                             Robust.sem
```

```
##   Information                                      Expected
##   Information saturated (h1) model              Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Confidence =~
##     V30               0.643    0.061   10.527    0.000    0.643    0.422
##     V15               0.764    0.048   15.790    0.000    0.764    0.545
##     V19               0.673    0.046   14.644    0.000    0.673    0.520
##     V24               0.747    0.036   20.631    0.000    0.747    0.634
##     V27               0.720    0.048   15.129    0.000    0.720    0.563
##     V21               0.703    0.039   18.050    0.000    0.703    0.590
##     V9                0.678    0.041   16.344    0.000    0.678    0.616
##     V6                0.635    0.046   13.919    0.000    0.635    0.529
##   Connection =~
##     V12               0.879    0.061   14.496    0.000    0.879    0.664
##     V5                0.794    0.059   13.440    0.000    0.794    0.571
##     V28               0.827    0.061   13.452    0.000    0.827    0.597
##     V13               0.502    0.058    8.584    0.000    0.502    0.469
##     V16               0.812    0.060   13.596    0.000    0.812    0.496
##   Uncertainty =~
##     V7                0.914    0.059   15.419    0.000    0.914    0.575
##     V10               0.738    0.057   13.010    0.000    0.738    0.463
##     V22               0.808    0.054   15.046    0.000    0.808    0.535
##     V25               0.546    0.064    8.478    0.000    0.546    0.333
##     V8                0.943    0.056   16.775    0.000    0.943    0.582
##     V4                0.699    0.053   13.264    0.000    0.699    0.499
##     V2                0.509    0.047   10.775    0.000    0.509    0.389
##   Empathy =~
##     V11               0.505    0.054    9.312    0.000    0.505    0.375
##     V26               0.662    0.059   11.232    0.000    0.662    0.491
##     V17               0.672    0.059   11.488    0.000    0.672    0.489
##     V1                0.680    0.057   12.000    0.000    0.680    0.484
##     V23               0.411    0.060    6.833    0.000    0.411    0.278
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Confidence ~~
##     Connection        0.596    0.045   13.362    0.000    0.596    0.596
##     Uncertainty       0.606    0.035   17.384    0.000    0.606    0.606
##     Empathy           0.741    0.039   18.931    0.000    0.741    0.741
##   Connection ~~
##     Uncertainty       0.668    0.037   17.907    0.000    0.668    0.668
##     Empathy           0.452    0.051    8.788    0.000    0.452    0.452
##   Uncertainty ~~
```

```
##     Empathy           0.475    0.052    9.213    0.000    0.475    0.475
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .V30               1.904    0.102   18.693    0.000    1.904    0.822
##    .V15               1.379    0.081   17.077    0.000    1.379    0.703
##    .V19               1.222    0.089   13.746    0.000    1.222    0.729
##    .V24               0.830    0.053   15.536    0.000    0.830    0.598
##    .V27               1.116    0.068   16.341    0.000    1.116    0.683
##    .V21               0.927    0.061   15.297    0.000    0.927    0.652
##    .V9                0.752    0.073   10.242    0.000    0.752    0.621
##    .V6                1.040    0.098   10.622    0.000    1.040    0.720
##    .V12               0.977    0.112    8.702    0.000    0.977    0.558
##    .V5                1.304    0.154    8.463    0.000    1.304    0.674
##    .V28               1.232    0.121   10.211    0.000    1.232    0.643
##    .V13               0.892    0.121    7.378    0.000    0.892    0.780
##    .V16               2.014    0.120   16.757    0.000    2.014    0.754
##    .V7                1.690    0.109   15.494    0.000    1.690    0.669
##    .V10               2.000    0.098   20.476    0.000    2.000    0.786
##    .V22               1.629    0.091   17.963    0.000    1.629    0.714
##    .V25               2.393    0.104   22.999    0.000    2.393    0.889
##    .V8                1.734    0.130   13.378    0.000    1.734    0.661
##    .V4                1.472    0.079   18.671    0.000    1.472    0.751
##    .V2                1.459    0.098   14.888    0.000    1.459    0.849
##    .V11               1.563    0.089   17.520    0.000    1.563    0.860
##    .V26               1.378    0.096   14.392    0.000    1.378    0.759
##    .V17               1.438    0.106   13.547    0.000    1.438    0.761
##    .V1                1.509    0.086   17.555    0.000    1.509    0.765
##    .V23               2.019    0.090   22.503    0.000    2.019    0.923
##     Confidence        1.000                               1.000    1.000
##     Connection        1.000                               1.000    1.000
##     Uncertainty       1.000                               1.000    1.000
##     Empathy           1.000                               1.000    1.000
```

```r
# Print only selected fit measures
fitMeasures(model.fit, c("cfi","tli", "rmsea", "srmr"))
```

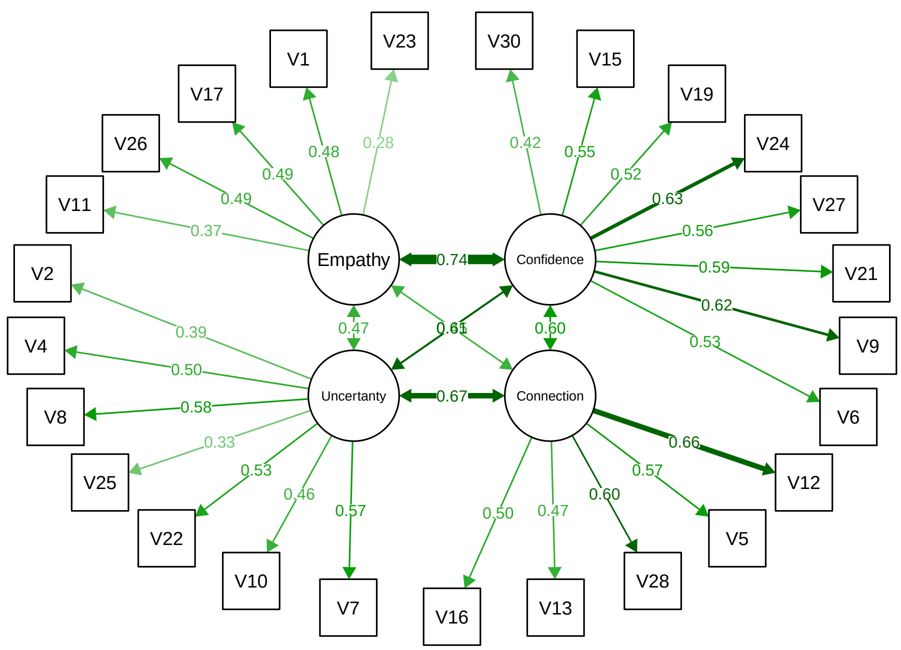
```
##   cfi   tli rmsea  srmr
## 0.818 0.797 0.058 0.056
```

The fit of our four-factor model resulted in CFI = 0.818, TLI = 0.797, RMSEA = 0.058, and SRMR = 0.054. Although some of the results like CFI or TLI are below the generally accepted threshold of 0.90, point at weak fit, one should be careful when relying on the CFA threshold values. As a final step of the CFA, we used *semPaths()* command from the

semPath package in R which can be used to visualize normalized values for all items and corresponding four factors.

Visualizing the normalized values for all items and corresponding factors

```r
semPaths(model.fit,
         what = "std",
         layout='circle2',
         intercepts=FALSE,
         residuals=FALSE,
         nCharNodes=10)
```



validity and Reliability Statistics

```r
reliability(model.fit)
```

```
##        Confidence Connection Uncertainty   Empathy
```

```
## alpha   0.7730767  0.6835493   0.6807308 0.5261348
## omega   0.7714726  0.6936940   0.6823795 0.5205996
## omega2  0.7714726  0.6936940   0.6823795 0.5205996
## omega3  0.7672490  0.6996599   0.6776520 0.5134824
## avevar  0.2975713  0.3182537   0.2427188 0.1834691
```

```
htmt(model = cfa.model,
     data = teiq)
```

```
##             Cnfdnc Cnnctn Uncrtn Empthy
## Confidence   1.000
## Connection   0.577  1.000
## Uncertainty  0.588  0.644  1.000
## Empathy      0.660  0.327  0.415  1.000
```